

Protocol and data collection for walks

Before the walk

- Note weather conditions (temperature, rain/no rain)
- Note starting location (as precise as possible, maybe take a screenshot of google maps with your current location)
- Make sure pressure map on connection screen is responding accurately

During the walk

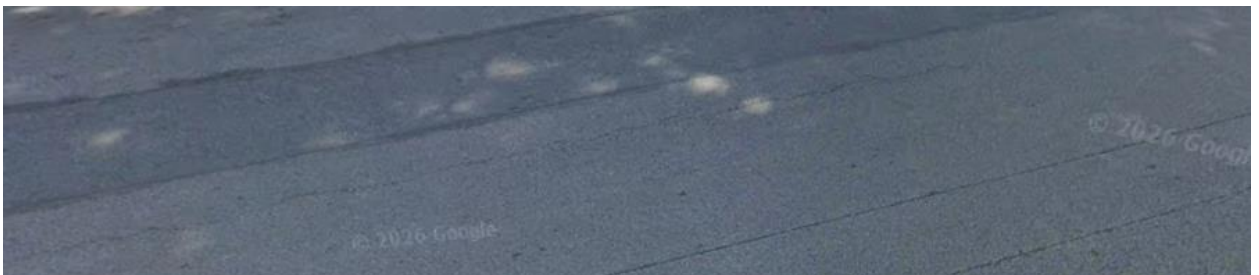
- An ideal walk is about 3 to 5 minutes, with multiple surfaces
- To the best of your ability, avoid stepping on sewers, drains, manhole covers, and other manmade surfaces (such as the yellow textured pads on either side of crosswalks)
- The walk itself should not be constantly switching surfaces every other step, as it makes it hard to tag and eliminates a lot of steps from the dataset. When doing a multi-surface walk, try to stay on the same surface for enough time so that it's not a handful of steps

After the walk

- In notes, add info from above, shoe model worn, ending location
- Make note of any times you stopped walking or any irregularities

Identifying surfaces

- Asphalt: often black, darker than concrete, most often found on roads, NOT usually sidewalks or pedestrian paths, but sometimes bike paths



- Concrete: any hard, paved surface that does not fall into asphalt. This covers most paved surfaces on sidewalks. Exposed aggregate falls into this category as well, but do not use that tag, we are dropping it



- Brick: self-explanatory, but there are multiple types of brick. We want to add brick that is not evenly paved to our dataset
- Grass: self-explanatory
- Dirt: natural, packed ground that does not have any man-made additions on top of it (i.e. no added stones, no added gravel)
- Compact ground: any natural surface that does not fall into bare dirt. Most commonly consists of light gravel and dust on top of dirt



Data we need collected:

Current step counts by user and material:

- Kristian: 5902
 - Grass: 1760
 - Loose: 893
 - Paved: 3249
- Sam: 6153
 - Grass: 1449
 - Loose: 1053
 - Paved: 3651

We want to try to have Samantha's dataset grow similar to Sam and Kristian, and we would like to have her add some more loose surfaces (gravel, crushed rocks, dirt, sand) as well as uneven brick, similar to cobblestone. The uneven brick falls under the 'paved' category, meaning the paved step count should include a decent amount of uneven brick, hence its higher number than Sam and Kristian. Here are the step counts we would like Samantha to reach:

- Samantha: ~7250
 - Grass: ~1500
 - Loose: ~1250
 - Paved: ~4500

At an estimated cadence of around 115 steps per minute, it breaks down to:

- Samantha: ~63 minutes
 - Grass: ~13 minutes
 - Loose: ~11 minutes
 - Paved: ~39 minutes

Sessions should not need to be more than 3-5 minutes in length, but should try to include as much variety in surfaces as possible, but should not be changing every few steps. Ideally it would be 2-3 clean sections.